

Aditya Raj Verma

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Education

Dr. B R Ambedkar National Institute of Technology Jalandhar

B.Tech in Electronics and Communication Engineering

Punjab, India

2024 – 2028

Certifications

AI Foundations – OpenAI

June 2026

Skills Summary

Languages: Python, C/C++

ML & Data: scikit-learn (Linear/Random Forest/Gradient Boosting, quantile regression), XGBoost, Hugging Face Transformers (NLP/sentiment analysis), Pandas, NumPy

Web & Deployment: Flask, FastAPI, Pydantic, HTML, CSS, JavaScript, Git/GitHub, Render, Vercel

CS Fundamentals: Data Structures & Algorithms (C++)

Core Electronics & Tools: Analog Circuits, Digital Communication, Signals & Systems, SQL, MATLAB, LTSpice

Experience

AI Intern

Jodhpur, India

Anand Rath IT Pvt Ltd (ARITPL)

June 2026 – August 2026

- Designed and shipped a solo, end-to-end mutual fund recommendation engine, scaling from 6 hand-picked funds to a 138-fund dataset spanning 7 categories using real AMFI NAV data
- Iterated through Linear Regression and Random Forest baselines, diagnosed an uninformative modeling target, and reframed the problem around quantile regression (GradientBoostingRegressor) to directly estimate downside risk; benchmarked against XGBoost
- Identified and resolved real data-quality defects in third-party financial data through statistical validation, including a corrupted NAV entry inflating returns by 100x and a systematic historical-sampling bias affecting 46% of the fund universe
- Integrated a live news-sentiment feature (Hugging Face FinBERT) and deployed the full system as a public Flask web application

Technical Workshops

- Attended 3-day Drone Bootcamp with hands-on drone assembly and flight systems
- Built RC car in robotics workshop; learned motors and control systems

Leadership & Community

- Led logistics for major college events (Utkansh, Arambh, Bharat Dhvani) as member of Main Event Organizing Club
- Managed online gaming community events and collaborations with industry partners (Jio Games, Krafton, Riot Games, Monster, Red Bull)

Projects

Investment Advisor ML | Python, Pandas, scikit-learn, XGBoost, Transformers, Flask, Render

- Built and deployed a full-stack ML recommendation engine mapping (principal, horizon, risk tolerance) to a risk-adjusted mutual fund recommendation, with best-case/worst-case/balanced return projections
- Engineered a 138-fund dataset across 7 categories from raw AMFI NAV data; found and fixed real data defects (corrupted NAV scaling, short-history sampling bias) through hypothesis-driven statistical investigation
- Trained and validated a quantile regression model (GradientBoostingRegressor) to estimate category-level downside risk; benchmarked against XGBoost with a rigorous MAE-based comparison
- Added a market-sentiment feature via live news retrieval and FinBERT sentiment scoring; deployed the full system as a public web app on Render

Health Insurance Price Estimator | Python, FastAPI, Pydantic, scikit-learn, JavaScript, Vercel

- Built and deployed a full-stack ML web app estimating health insurance costs from a Random Forest Regressor, with a FastAPI backend and vanilla JS/HTML/CSS frontend
- Designed REST endpoints with Pydantic request validation, connected a serialized (joblib) model to the API, and handled CORS between frontend and backend
- Deployed as a serverless FastAPI application on Vercel with auto-generated Swagger/OpenAPI documentation

Power Supply with Load Regulation & IoT-Based Monitoring

- Integrated voltage and current sensors with ESP32 for wireless data transmission and cloud-based IoT monitoring via ThingSpeak
- Developed real-time remote visualization dashboard for live sensor data
- Designed regulated DC supply (230V AC to stable DC) with load regulation to maintain stable output under varying load

Relevant Coursework

Data Structures & Algorithms, DBMS, Probability & Stochastic Processes, Analog Integrated Circuits, Digital Communication Systems, VLSI

Additional

Solved 300+ problems on LeetCode

Actively strengthening skills in Python & ML